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Biosensors exploit the molecular recognition properties that are unique to biomolecules to offer highly specific assays for compounds in complex mixtures. Many common biosensor designs are prone to significant mass transport limitations that lead to prolonged sensor response times. As a result, such biosensors have limited use in real-time monitoring of dynamic processes.
This research aims to accelerate mass transport in two types of barrier regions that occur in common biosensor designs. For biosensors based on gel-entrapped enzymes, we are developing an “internal pumping” method, wherein charged colloids are entrapped in the gel along with the enzymes used for sensing, and application of an external electric field drives electroosmotic fluid flow around the particles. Chaotic advection will be driven by applying a randomly fluctuating quadrupolar electric field that drives temporally and spatially random electroosmotic flows. Chaotic advection is an effective tool to promote mass transfer in laminar flows. For biosensors that require a membrane to separate the active sensing agents from the test solution, we will develop an “interfacial micromixing” technique,

Task Description:	wherein a dense array of charged, biofunctionalized colloids or live cells is deposited by electrophoretic deposition onto the surface of a microporous membrane. An externally applied electric field drives strong recirculating flows around the particles or cells where the biorecognition process occurs. In addition, similar electrophoretic deposition techniques are being used to produce patterned arrays of living cells and cell/colloid mixed arrays as a demonstration of concept for novel biosensors. Complementary to the main objective of this project, this latter task may also improve efforts to culture organized arrays of multiple cell types to engineer multicellular tissues, and it also should lead to new technologies for fractionating cell mixtures according to differences in their dielectric properties. The experimental program includes optical and spectroscopic techniques to measure mass transport rates in gels and on membrane surfaces during the application of the internal pumping and interfacial micromixing. The corresponding enhancement of reaction rates is being tested using the enzyme beta-galactosidase. Here the biosensor response time is dictated by the chemical reaction rate. In order to optimize these electrokinetic mixing techniques, the experimental plan is being complemented by theoretical analysis of electrophoretic and electrokinetic mass transfer in gels with simultaneous chemical reactions. The resulting predictions will be used to select operating conditions for maximum response rates. The objective of this research program is to develop electrokinetic techniques that minimize mass transport limitations in biosensors and thereby improve their signal response times. This is intended to expand the applicability of biosensors to dynamic processes with short timescales. Two techniques, “electroosmotic internal pumping” and “interfacial micromixing” will be developed experimentally and optimized with the guidance of a complementary theoretical investigation of electrokinetic flows around particles confined within gels or on a membrane wall. This research is developing techniques to increase the speed at which biosensors detect molecules in solution. Biosensors contribute significantly to the approximately $19 billion world market for medical diagnostics. Biosensors use biological molecules to detect trace amounts of specific chemical compounds in complicated chemical mixtures, for example, detecting hormones or sugars in blood, or toxins in groundwater or processed foods. A major limitation of biosensors is that they often require long times to produce a stable signal, referred to as a long response time. The long response time problem prevents biosensors from being used in any situation where chemical concentrations change rapidly, for example preventing one from using biosensors to monitor rapidly changing product concentrations in a pharmaceutical manufacturing process. Furthermore, at a time when combinatorial chemistry is allowing thousands of new molecules to be generated and tested for pharmaceutical activity, there is a need for rapid analysis, or “high throughput screening” of the biological effectiveness of those molecules. Biosensors are ideally suited for measuring biological effects, but the long response time problem significantly limits their use in a high throughput screening mode. The results of this research program are expected to expand the economic market for biosensors and to accelerate the screening of new pharmaceuticals in order to deliver new therapies to the general population. Furthermore, introducing new methods to fractionate cell mixtures should facilitate medical diagnostics.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Developing techniques to increase biosensor response rates will expand the number and type of industrial and biotechnological processes that can be monitored effectively by biosensors. For example, faster response times will increase the productivity of biosensor-based high throughput analyses of pharmaceutical candidates. In addition, the development of techniques to manipulate and fractionate cell mixtures may lead to new medical diagnostics, perhaps including disease diagnoses based on altered cellular dielectric properties.
Task Progress:	1. Internal Pumping. Our earlier experiments that demonstrated electroosmotically enhanced mass transfer in gels by doping them with highly charged particles and applying transverse electric fields were qualitatively, but not quantitatively, reproducible. We identified problems with non-uniform particle distributions in the gels, mechanical instabilities during electric field application and electrochemical reactions of buffer components. Following extensive refinements of our gel synthesis and analytical procedures, experimental results are now extremely reproducible. This has enabled systematic studies of the effects of fields strength, particle concentration and particle size on the degree of mass transfer enhancement. The enhancement scales with the total surface area of embedded particles, as expected for a surface-driven flow process. Using dc fields aligned parallel to the solute concentration gradient, solute fluxes are enhanced by electroosmosis. For example, at a particle load of 2 vol%, solute flux is doubled by application of a 3 mA current across the gel. If the field is applied antiparallel to the solute concentration gradient, it is possible to drive an electroosmotic flow that runs counter to diffusion and prevents solute flux across the gel. We anticipate that this may become useful as an electrically controlled microfluidic valve or as an electrically controlled drug delivery technology. For example, medication could be released on demand from an implanted gel reservoir when needed, in response to an external electrical signal. 2. Interfacial Micromixing and Particle Manipulation. We have demonstrated that cells and particles can be differentially manipulated and positioned adjacent to a microelectrode array when an ac field is applied between the array and a uniform electrode oriented parallel to the array. Cells and particles migrate in response to a coupling between ac electroosmosis and dielectrophoresis. Thus, the separation is based on differences in cell or particle polarizability. This has been exploited to efficiently separate different particle types at low frequencies (~10 – 1000 Hz) and low electric field strengths (~ 10 V/cm) that are appropriate for biological applications. Separations were first demonstrated in a model system consisting of yeast and colloidal polystyrene. Yeast has also been separated from red blood cells using this technique. This experimental work has been complemented by theoretical analysis. We have developed a new model of particle polarizability derived from the full electrokinetic theory, as well as a new model of frequency-dependent electrode impedance and the structure of ac electric fields in conducting media. These theoretical analyses indicate that cells with small differences in dielectric properties can be separated by judiciously specifying the conductivity of the suspending medium. This variable is key because it not only affects how the difference between two cells’ polarizabilities varies with frequency, but it also controls the electric field structure around the microelectrodes. We anticipate that cells with small differences in membrane conductivity, perhaps because of a disease state, could be separated by this technique. We have also demonstrated the ability to position vesicles using this technique. Vesicles can patterned and subsequently ruptured by the electric field, providing a high degree of spatial control over vesicle content release behaviors. Vesicles can be loaded with reactants, and by differentially manipulating different reactant-laden vesicles or colloidal particles, we can control when and where reactions occur in microfludic devices.

Bibliography Type:	Description: (Last Updated: 08/30/2006)
Articles in Peer-reviewed Journals	Zhou, H.; White, L.R.; Tilton, R.D. "Lateral Separation of Colloids or Cells by Dielectrophoresis Augmented by AC Electroosmosis" Journal of Colloid and Interface Science in press , Dec-2004
Articles in Peer-reviewed Journals	Zhou, H.; Preston, M.A.; Tilton, R.D.; White, L.R. "Calculation of the electric polarizability of a charged spherical" Journal of Colloid and Interface Science in press , Jan-2005